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Aerial RF and Throughput Measurements on a Non-Standalone 5G Wireless Network

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***Abstract*—In this paper, we report experimental results from aerial measurements on a 5G non-standalone (NSA) network in C-band using a 5G-enabled unmanned aerial vehicle (UAV). As the demand for using advanced communications to support various use cases of UAVs rapidly increases, understanding the performance of 5G terrestrial networks in the 3D spatial domain becomes critical. Characterizing the 3D propagation between terrestrial networks and UAVs also reveals spectrum reuse opportunities between terrestrial and non-terrestrial networks. In this paper, we compare altitude-dependent radio frequency (RF) signal characteristics in the sub-6 GHz bands, specifically in the C-band measured by a 5G-powered aerial platform. In particular, we analyze the field-measured RF signal parameters of a 5G non-standalone (NSA) network with both LTE and NR radios deployed in the AERPAW test bed. We discuss how the RF and throughput measurements vary with link distance and 3D location, and their implications on 3D spectrum reuse.**

***Index Terms*—4G, 5G, NSA, LTE, NR, AERPAW, C-band, UAV, RSRP, SINR, IPerf, air-to-ground, software-defined radio.**

I. Introduction

With the ongoing spectrum strategy to cater to the demand for spectrum access [1], innovations in wireless technologies are growing rapidly. Both in the government and private sectors, next-generation Wi-Fi networks, satellite systems, 5G and 6G broadband networks, private wireless networks, autonomous vehicles, and other advanced systems are driving the demand. Aerial communications, device-to-device (D2D) satellite services, and dynamic spectrum sharing are some of the key areas of recent focus to enhance the 5G and beyond.

In order to be able to meet the ever-increasing spectrum demand, it is critical to reuse the available spectrum resources. Due to their favorable propagation characteristics, spectrum reuse in the sub-6 GHz band received the heaviest attention, including the 3100-3450 MHz and 5030-5091 MHz bands specified in the national spectrum strategy [1], both of which involve aerial transmitters. The 3D radiation and signal propagation conditions in these bands are not studied for unmanned aerial vehicles (UAVs), and include intricate phenomena, such as the UAVs being served by sidelobes of cellular towers at high altitudes [2]. These irregularities may result in frequent handovers between serving base stations (BS) [3], while they may also introduce unique spectrum reuse opportunities [2], [4], [5]. For example, when a UAV is at the null of a 5G antenna pattern, that UAV may reuse the same spectrum while the BS is serving another ground user.

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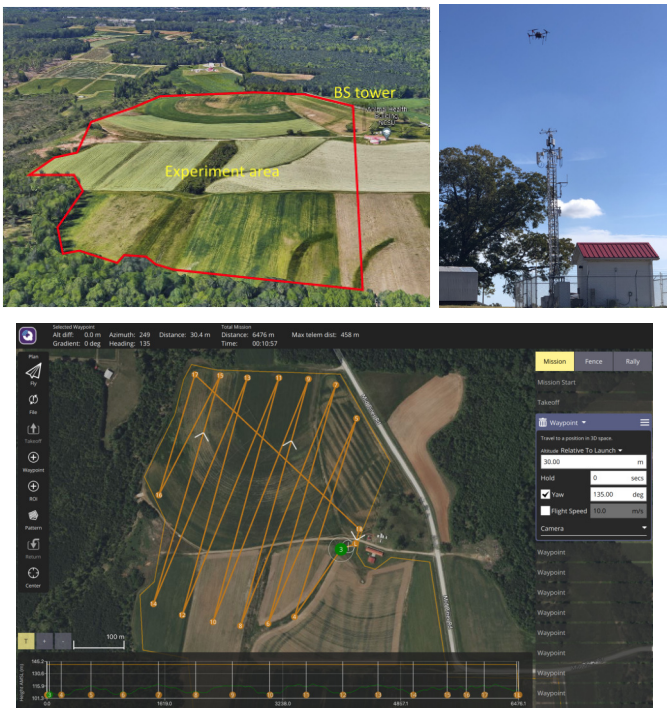

Fig. 1. UAV experiment area in AERPAW (top-left), 5G NSA Ericsson tower (top-right), and mission plan for a sample UAV experiment (bottom).

In this short paper, we present air-to-ground propagation measurements from the NSF AERPAW platform located at NC State University (see Fig. 1) using a non-standalone (NSA) 5G network operating at the C-band. The rest of the paper is organized as follows. Section II overviews the experiment scenario, Section III presents the radio frequency (RF) and throughput measurements and related discussions, and the last section concludes the paper.

II. Experimentation Description

AERPAW testbed constitutes various types of wireless radio and access platforms of 4G/5G, RF sensors, UAVs etc. The AERPAW infrastructure is predominately spread across two main locations: one at the Centennial campus of NC State University which gives a more of an urban environment, and a second one at the Lake Wheeler site which gives a rural and agricultural area type of environment. Both these locations are equipped with radio infrastructure providing 4G and 5G coverage. One of them is a commercial grade 5G network based on Ericsson equipment which is the main focus of the

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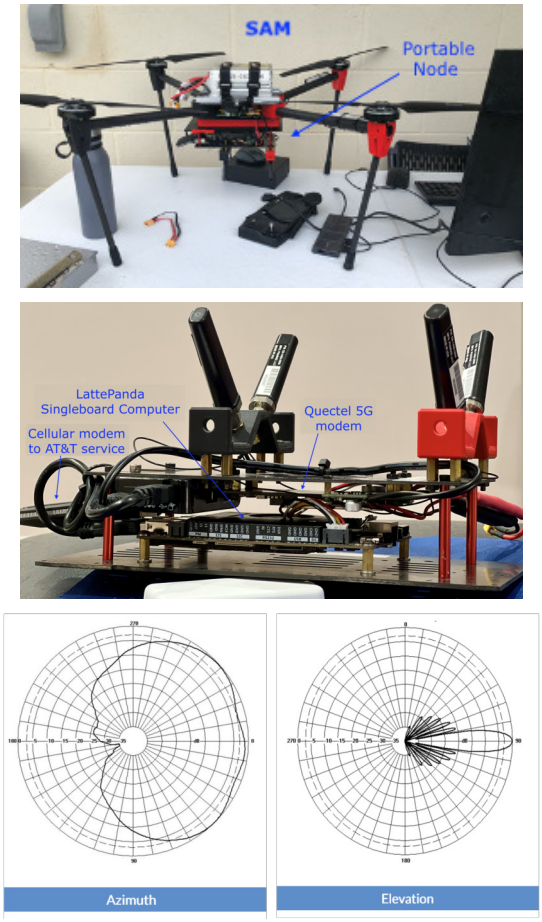


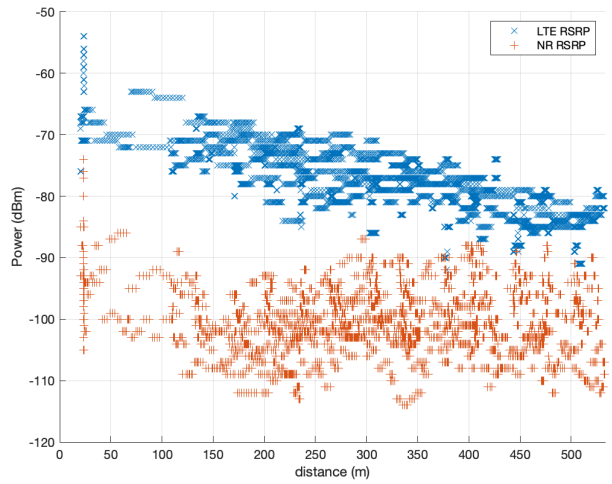
Fig. 2. Small AERPAW multicopter (SAM) (top) mounted with a portable radio node (middle) used in air-to-ground measurements, and the NR sector 3D antenna radiation pattern at the gNB (bottom).

experiment for this study. The RAN of this 5G system is deployed at the AERPAW Lake Wheeler site which is a rural agricultural area, predominantly a open aerial field with some vegetation on the ground as shown in Fig. 1.

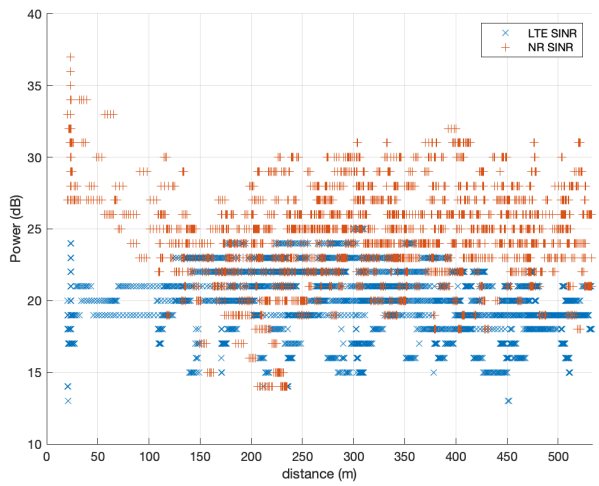
To characterize the aerial performance of a 5G system, we used a 5G NSA system with overlaid NR and LTE sectors. The LTE anchor carrier is in band 66 (1.7/2.1 GHz) with 5 MHz channel bandwidth and an NR carrier at 3.4 GHz in band n77 with 100 MHz of channel bandwidth. The sectors use a pair of dual +/-45 deg cross polarized directional antennas with 120 degrees of azimuth beam width (Fig. 2) facing the north-west direction from the base station tower. LTE uses a 2x2 MIMO on the downlink whereas for NR it can be set to either 2x2 or 4x4 MIMO. For this experiment, both LTE and NR carriers are set to 5 watts of transmit power per antenna port.

For aerial measurements, we used a portable node as a 5G device carried by a UAV. The portable node uses a Quectel 5G modem module connecting to the Ericsson 5G base station, a LattePanda module as a companion computer to interface with the UAV, and an ATT 5G modem for command and control. This portable node is mounted on a small AERPAW Multi-rotor (SAM) UAV as shown in Fig. 2.

Fig. 1 also includes the flight mission plan that was used on the UAV for this experiment. It is a zigzag aerial path covering the north-west quadrant of the area from the base station tower



(a) RSRP of the LTE and NR carriers.



(b) SINR of the LTE and NR carriers.

Fig. 3. RSRP and SINR of LTE and NR with respect to distance.

as shown in the figure. For this experiment, the measurements are done at a constant altitude of 30 m and speed of 10 m/s after the flight took off, and the yaw of the flight was kept constant as well. The distance of the flight from the BS tower can be seen from the later figures that the odd peaks with larger distances correspond to the left edges and even peaks correspond to the right edges of this mission plan.

III. Measurement Results

As explained before we used a UAV powered by a 5G device, a Quectel modem in this case, to measure the parameters and collect data along a zigzag aerial path trajectory given in Fig. 1. With this experiment two types of measurements are collected for analysis: a) RF parameters like Reference Signal Receive Power (RSRP) and Signal-to-interference and noise ratio (SINR) and b) Application layer performance parameter like throughput.

A. RF Measurements

RSRP and SINR are some of the RF level parameters measured at the 5G-powered UAV along with the flight geo-

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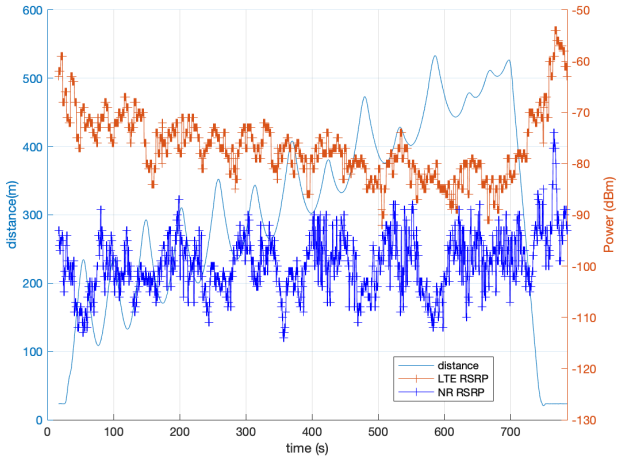
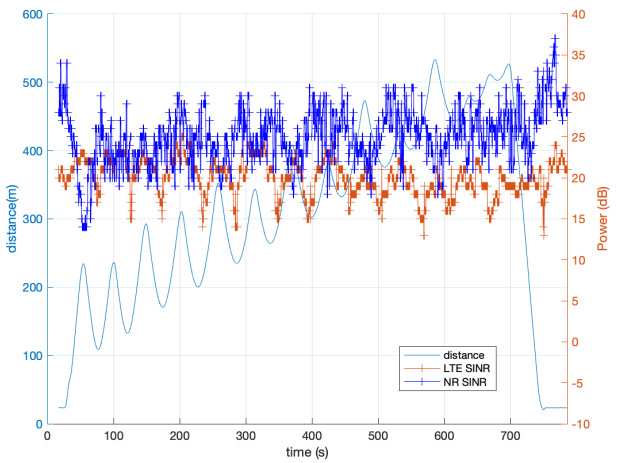
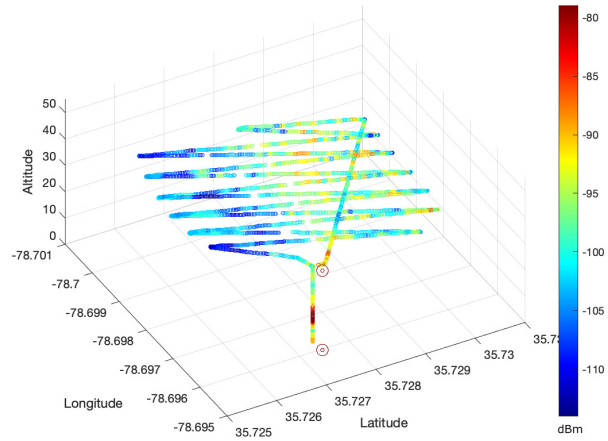


(a) RSRP of the LTE and NR carriers.

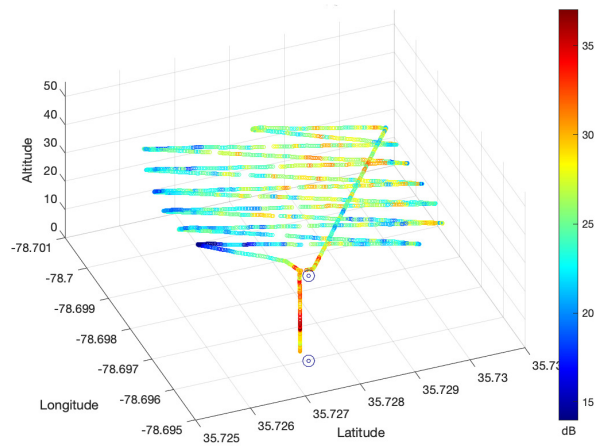


(b) SINR of the LTE and NR carriers.

Fig. 4. RSRP and SINR of LTE and NR with respect to distance and time.



(a) RSRP of the NR carrier.



(b) SINR of the NR carrier.

Fig. 5. RSRP and SINR of NR with respect to UAV location.

locations and time. Fig. 3(a) and Fig. 3(b) give the measured RSRP and SINR levels on the LTE and NR carriers versus distance, respectively. From Fig. 3(a), the RSRP levels on the NR carrier seem to be lower than that of the LTE carrier over the measurement distance, but on the contrary, the SINR on the NR carrier from Fig. 3(b) show a higher level than that is on the LTE carrier over the same distances. Another observation from Fig. 3(a) is that the NR carrier RSRP level is less prone to the distance over the measurement distance of 530 m from this experiment, and is evenly distributed between -90 dBm and -110 dBm across the measurement distance, whereas the LTE carrier RSRP level seems to be decreasing with distance.

The RSRP and SINR vs distance and time as in Fig. 4(a) and Fig. 4(b) give a different perspective of these RF parameters. These plots again show the RSRP level on LTE carrier is stronger than NR carrier and SINR level on the NR carrier is higher than the LTE carrier. The distance vs time (blue line) from these plots corresponds to the vehicle zigzag path, and the larger (odd numbered) peaks on this line correspond to the left-most points and the smaller (even numbered) peaks correspond to the right-most points of the vehicle zigzag path trajectory. From Fig. 4(a) it is evident that the RSRP level in the leftmost side of the mission plan is lower, say about -110 dBm whereas on the right half side of the trajectory is about 15 dB higher, and this must be from the beam width of the directional antenna which is pointing towards the top-right corner (NW direction in the field). Similarly the same can be observed for SINR as well from Fig. 4(b). This is in fact more evident from Fig. 5(a) and Fig. 5(b) which shows the RSRP and SINR levels are lower on the left side of the mission plan that corresponds to the edges of the main lobe of the directional antenna that is pointing more towards the right side, i.e., towards the NW direction.

B. Throughput Measurements

At the application layer we used IPerf traffic throughout the flight of the UAV to measure the downlink throughput performance. Fig. 6 and Fig. 7 give the traffic performance results measured from this experiment. From these, it can be seen that a) the maximum throughput achieved was 340 Mbps, and b) a drop in the throughput at the left edges of the flight path which is visible in Fig. 6(b), i.e., the odd-numbered peaks

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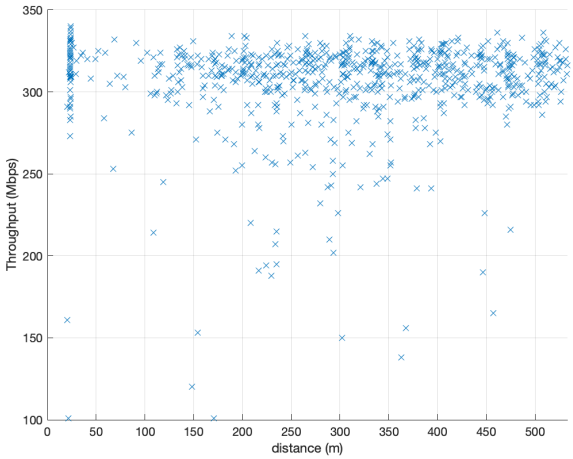
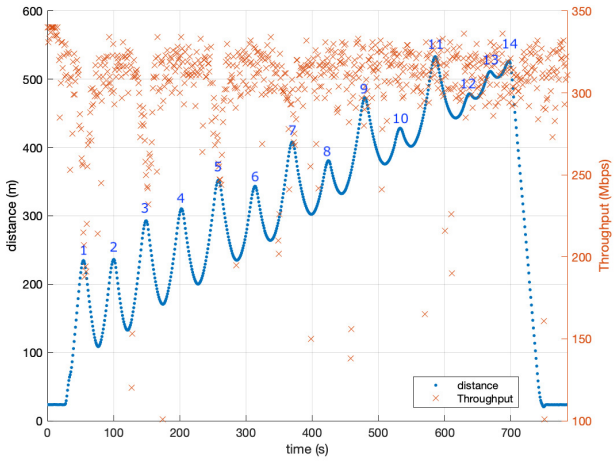
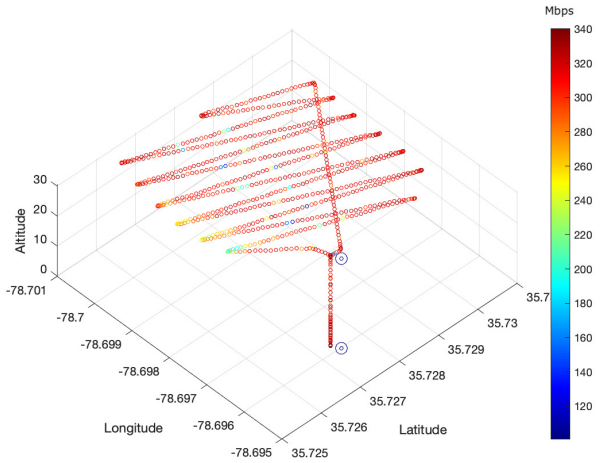


(a) Throughput versus distance.

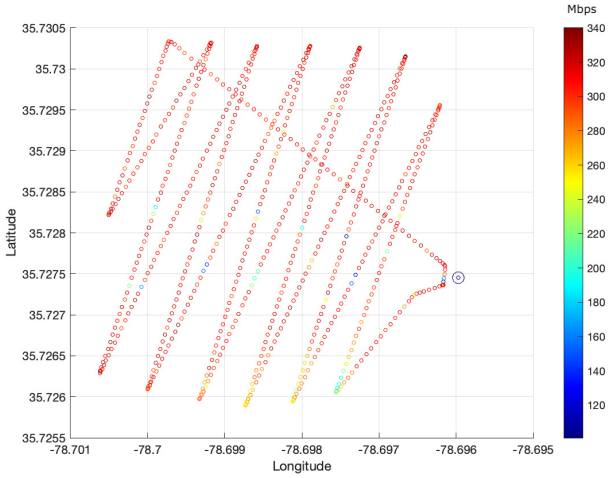


(b) Throughput versus distance and time.

Fig. 6. Throughput versus distance and time.



(a) Throughput versus 3D location.



(b) Throughput versus 2D location.

Fig. 7. Throughput versus 3D and 2D location of the UAV.

from the distance versus time plot in Fig. 6(b) correspond to the left edges of the fight path. This observation aligns with the drop in the RSRP and SINR measurements as well. Otherwise, the majority of the time, the throughput stays above 300 Mbps across the entire distance and time of the flight.

IV. Conclusion and Discussion

We presented air-to-ground RF propagation and throughput measurements on a non-standalone (NSA) 5G network operating at the C-band. From the results we observe that the NR signal in C-band radio frequency seems to be more immune to the aerial environment over the measurement field but it may be more dependent on the 3D radiation pattern of the antenna instead. This paper also highlights the capabilities of the NSF AERPAW platform at the NC State University for researchers in academia and the industry and the ongoing works in these areas. With the ongoing national spectrum strategy studies, especially for the aerial use cases, these experiments underline the need for further measurements and insights on the performance of 5G in 3D aerial scenarios.

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